

Product Feature Specification Proposal: B2B Beverage Distribution & Sales Platform

1. Core Pillar: Commercial Calculation & Logistics Engine

Unlike standard consumer storefronts that process static, single-item transactions, this engine automatically derives commercial supply chain logistics, multi-jurisdictional financial parameters, predictive analytics, and delivery timelines based on systemic trigger events and strict business-to-business (B2B) operational constraints.

A. Dynamic Volume & Tiered Pricing Algorithm

- **Contract-Driven Inventory Scaling:** Engineered to monitor wholesale ordering thresholds, programmatically altering the baseline item cost in real time based on volume metrics.
- **Tiered Price Depreciation:** The logic automatically calculates unit cost depreciation matrices based on order size (e.g., standard wholesale rate for 1–9 cases; automatic percentage discount triggers when an order reaches a 10+ case threshold).
- **Automated Parameter Verification:** Employs built-in checks to ensure total order quantities comply with minimum order quantities (MOQ) established for specialized inventory classifications.

B. Logistical Lead-Time & Delivery Padding Rules

- **Fulfillment Window Optimization:** Employs a custom programmatic calculation to establish delivery arrival dates based on geographical retail regions and distribution center workloads:

$$D_{\text{delivery}} = D_{\text{order}} + P_{\text{transit}} + \text{Fulfillment Buffer}$$

- **Sovereign Calendar Filtering:** The algorithm references a localized calendar helper to automatically strip out weekends and statutory holidays, ensuring commercial freight windows reflect true operational business days.
- **Carrier Variable Mapping:** The transit variable (P_{transit}) automatically scales its padding parameters based on fulfillment distribution tracks, adding 5 days for standard cross-provincial freight or 2 days for localized regional courier routing.

C. Regional Tax & Multi-Jurisdictional Compliance

- **Ontario HST Computation Matrix:** Automatically appends localized taxation frameworks (such as Ontario's 13% Harmonized Sales Tax) directly onto wholesale itemization layouts.
- **Pre-Submission Financial Auditing:** Runs automated verification checks on subtotal payloads, ensuring tax exemptions (e.g., specific manufacturing or native status exemptions) map perfectly before an invoice payload is signed and locked.

D. Predictive Route Logistics & Demand Forecasting Engine

- **Historical Order Frequency Aggregation:** The application dynamically captures and aggregates data points from every finalized order payload. Over time, it maps these entries into an analytics dataset, tracking exact ordering intervals, recurring seasonal spikes, and client-specific volume frequencies.

- **Predictive Route Mapping Optimization:** Using an analytics logic module, the engine compiles the order frequency datasets across all active corporate accounts in a specific geographic region. It translates these consumption patterns into a predictive logistics footprint, mapping an anticipated territory routing layout for the upcoming week.
- **Pre-Fulfillment Fleet Planning Dashboard:** Provides distribution administrators with an interactive forecasting interface displaying a simulated visualization of next week's delivery routes. The system automatically estimates total required truck capacity, flags accounts with a high statistical probability of placing an order, and calculates total anticipated case volumes.
- **Proactive Inventory Pre-Allocation:** Ties the predictive routing output directly to warehouse management logic, alerting operators to pre-stage high-velocity SKUs onto physical picking layers before the automated re-ordering prompts are approved by the buyers.

2. Supporting Pillar: Advanced Distribution Management Features

These integrated features elevate the platform from a simple digital catalog into a robust enterprise operations tool, ensuring secure client operations, automated workflow tracking, and intelligent purchasing assistance.

A. Secure Account Authentication & Custom Portals

- **Role-Based Access Control (RBAC):** Restricts interface access based on organizational identities. Authorized corporate purchasers can execute orders, accounting staff can access financial transaction logs, and warehouse personnel manage inventory states.
- **Secure Client Intake Pipeline:** A secure onboarding channel that validates corporate registrations, retail sales licenses, and tax certificates prior to generating operational client credentials.
- **Session Security Architecture:** Implements persistent token-based authentication on physical iOS hardware devices, keeping enterprise data lines fully secure during background database operations.

B. Intent-Based Search & Directory Engine

- **Direct Name Lookup:** Features a real-time lookup bar anchored directly to the user interface, utilizing a localized predicate filter to isolate items instantly as the user types.
- **Multi-Tiered Category Filtering:** Segments the complete inventory master catalog into organized commercial columns (e.g., Juices, Energy Drinks, Bulk Pallets, Case Bundles) through strict data model categorizations.
- **Intersection Compound Logic:** Combines text entries and category tags via a compound boolean query string, matching product names, SKU IDs, or flavor profiles instantly without triggering full page reloads or network lag.

C. Automated Document & Invoice Assembly

- **Trigger-Based Assembly:** Links directly to the checkout engine to dynamically auto-assemble and populate official corporate logistics paperwork, outputting clean, un-filed copies of commercial invoices, picking slips, and bill of lading documents.
- **Smart Data Extraction:** Pulls stored client profiles, shipping addresses, tax identifiers, and customized contract pricing metrics directly from the local database to map text layers onto official PDF templates with flawless formatting.

D. Predictive Inventory & Automated Re-Ordering Prompt

- **Velocity-Based Consumption Tracking:** The system monitors a client's historical order frequency and volume metrics to establish an average product consumption velocity (e.g., calculating that a specific retail store goes through 5 cases of juice every 14 days).
- **Automated Re-Order Triggers:** When the calculated timeline approaches the end of a product's estimated lifecycle, the application generates a local system notification or proactive dashboard prompt reading: *"Running low on Sunberry Mango? Tap here to duplicate your last order."*
- **One-Tap Order Replication:** Clicking the re-order prompt automatically repopulates the client's shopping cart with their exact historical SKU quantities, applies their active contractual pricing tier, and moves them directly to the final confirmation screen.
- **Low-Stock Safety Thresholds:** Integrates a safety buffer logic module that coordinates with warehouse inventory levels to alert buyers to re-order popular seasonal or high-turnover product lines before a regional distribution shortage occurs.

E. Logistics Verification & Evidence Upload Engine

- **Digital Signatures & Intake Pipelines:** Integrates legally binding e-signature software within the delivery pipeline to confirm product receipt at retail drop-off zones.
- **Discrepancy Reporting Layer:** Provides a portal allowing warehouse operators or retail managers to capture and upload documentation (e.g., signed proof-of-delivery slips or photos of damaged inventory layers) directly into the file.
- **Automated SMS Reminders:** Sends text-message alerts to client point-of-contacts regarding distribution truck dispatch times, route tracking updates, or payment collection milestones.

3. Deployment Options

To address the diverse operational requirements of cloud-first distributors, localized logistics firms, and rapid shared-screen technical grading evaluations, the platform architecture accommodates three distinct deployment tracks:

- **Cloud Architecture:** Utilizing a secure, multi-tenant SaaS infrastructure hosted out of enterprise data centers located strictly in Canada Central (Toronto) to adhere to Canadian data residency principles and ensure low-latency inventory syncing.
- **On-Premises / Local Infrastructure Option:** A containerized deployment model allowing multi-partner distribution facilities or enterprise warehouses to install the application database directly on hardware located physically at the client's location. This enables physical data sovereignty and continuous offline inventory management, while maintaining a secure external cloud bridge to receive automated rule updates and pricing changes.
- **Local Sandbox Xcode Compilation (Capstone Development Track):** Designed explicitly to support local sandbox execution and rapid feature validation without production cloud dependency. The application compiles natively via Xcode to target the macOS iOS Simulator (ideal for shared-screen technical demonstrations and rapid UI evaluations) or side-loads directly onto a physical iOS hardware device via a local wired or wireless connection. This option maintains full local programmatic logic, core structural calculations, and offline-first database compliance tracking powered by SwiftData, with remote codebase synchronization managed through a clean Git/GitHub source tree.